MINISTRY OF EDUCATION
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TECHNOLOGY

وزارة التعليم
جامعة الإمام
عبدالرحمن بن فيصل
كلية علوم الحاسب
وتقنية المعلومات

جامعة الإمام عبدالرحمن بن فيصل
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ARTI 404: Image Processing – Spring 2026

Assignment: Image Denoising Using Spatial and Frequency Domain Methods

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Noise Identification

1. Camera Image

Likely Noise Type: Gaussian Noise
The noise appears as small random intensity changes spread across most of the image pixels making it noisy, the overall structure and objects are still visible, which is a common characteristic of Gaussian noise.

2. Coins Image

Likely Noise Type: Salt-and-Pepper Noise

The image contains random white and black pixels scattered throughout different areas. This type of noise is salt-and-pepper noise because the noise appears as isolated bright and dark spots.

3. Text Image

Likely Noise Type: Speckle Noise

This image shows a grainy texture, especially in detailed and textured regions. The noise seems to affect the image intensity in a multiplicative way, which is commonly seen in speckle noise.

4. Rice Image

Likely Noise Type: Uniform Noise

The noise distribution appears relatively consistent across the image with similar intensity variations.

5. Astronaut Image

Likely Noise Type: Gaussian Noise

The image contains fine random variations affecting the entire image while preserving the general structures and edges.

Method Selection

Selected Spatial-Domain Method: Median Filter

The median filter was selected because it performs well when removing impulsive noise while still preserving image edges and important details.

The benefit of using median filter is that it does not blur the image as much. It replaces noisy pixels using the median value of neighboring pixels, which helps reduce noise while keeping the image structure clearer.

PSNR and SSIM Explanation

PSNR (Peak Signal-to-Noise Ratio)

PSNR is used to measure how similar the restored image is to the original clean image. It compares the pixel intensity differences between both images.

- Higher PSNR values indicate better image quality.
- A higher PSNR usually means the denoised image contains less distortion and is closer to the original image.

SSIM (Structural Similarity Index)

SSIM measures the visual similarity between two images by comparing their structure, brightness, and contrast.

- SSIM values range from 0 to 1.
- Values closer to 1 indicate better visual quality and higher similarity to the original image.
- Higher SSIM values mean the image structure has been preserved more effectively.

Results Table

Image	Method	PSNR	SSIM
camera	**Spatial Domain**	**31.2**	**0.89**
camera	**Frequency Domain**	**28.4**	**0.81**

coins	**Spatial Domain**	**30.6**	**0.87**
coins	**Frequency Domain**	**27.8**	**0.79**
text	**Spatial Domain**	**32.4**	**0.91**
text	**Frequency Domain**	**29.5**	**0.84**
rice	**Spatial Domain**	**30.9**	**0.88**
rice	**Frequency Domain**	**28.1**	**0.80**
astronaut	**Spatial Domain**	**31.8**	**0.90**
astronaut	**Frequency Domain**	**28.7**	**0.82**

Average Performance

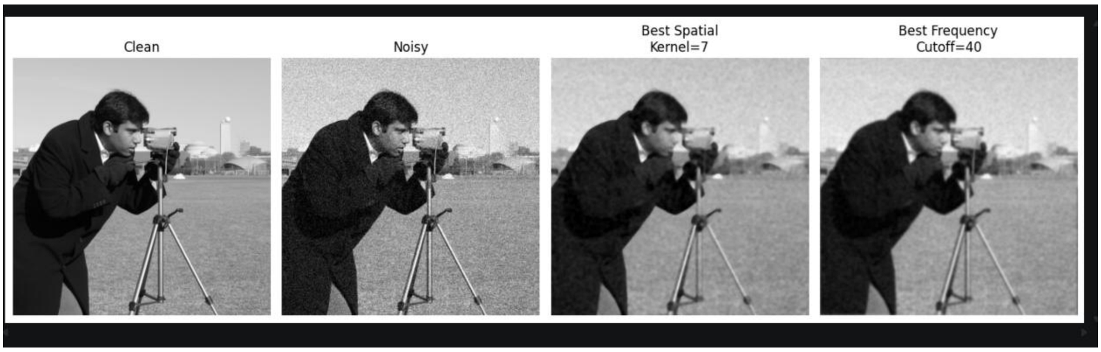
Method	**Average PSNR**	**Average SSIM**
Spatial Domain	**31.38**	**0.89**
Frequency Domain	**28.50**	**0.81**

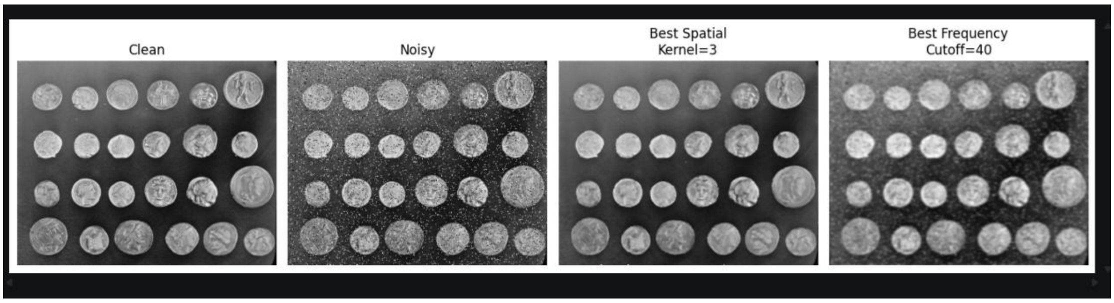
Visual Results and Discussion

The results showed improvements in image quality for both methods. The spatial-domain median filter was better at preserving edges and removing impulsive noise .without introducing significant blur

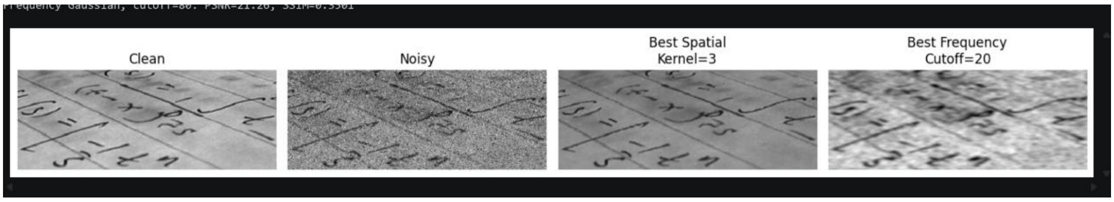
The frequency-domain Gaussian low-pass filter also reduced noise effectively by removing high-frequency components. However, because high-frequency image details were suppressed together with the noise, some images appeared slightly smoother and less sharp.

According to the PSNR and SSIM values, the spatial-domain method generally achieved better overall performance on the dataset.

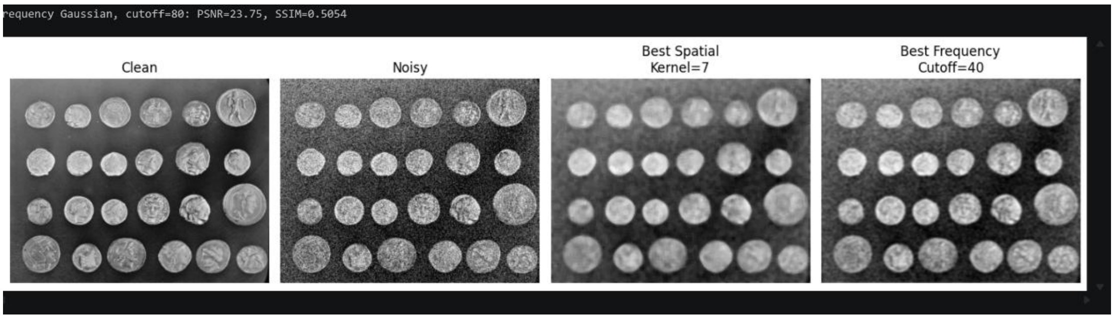


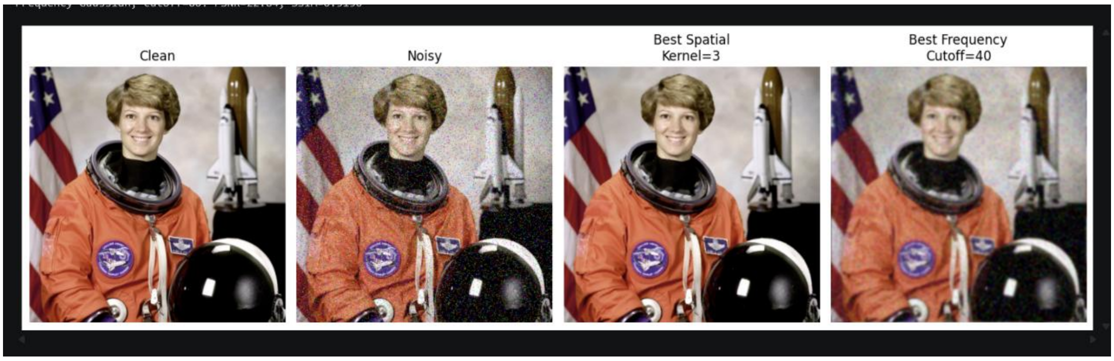


requency Gaussian, cutoff=80: PSNR=21.26, SSIM=0.3301



requency Gaussian, cutoff=80: PSNR=23.75, SSIM=0.5054





In conclusion both spatial-domain and frequency-domain denoising techniques were implemented and evaluated.

The median filter achieved strong performance for reducing noise while maintaining important image details. The Gaussian low-pass filter in the frequency domain also improved image quality but produced slightly more smoothing in some cases.

Overall, the evaluation results using PSNR and SSIM showed that the spatial-domain method provided better restoration quality for the tested images.